M15: A Multiscale Post-Hoc Protocol for Auditing the Geometric Fidelity of Latent Projections

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Abstract

Low-dimensional (2D) visualization of semantic latent spaces or influence networks is a critical tool for decision-making in analytical, forensic, and institutional contexts. However, these visual maps typically lack formal geometric fidelity guarantees. We present a multiscale post-hoc protocol to audit the geometric fidelity of 2D latent projections, combining three complementary layers: (i) the Mantel Test for global metric fidelity, (ii) a 0-dimensional persistence profile (**PH0 Profile**) over the Minimum Spanning Tree (MST) for hierarchical topological stability, and (iii) a *Fissure Index* based on community permutations to detect local semantic-structural dissonance. An adversarial benchmark of 1,000 Monte Carlo simulations (10 scenarios, $N = 100$ nodes) and a comparative ablation study against classical metrics (Venna & Kaski, 2001; Lee & Verleysen, 2007) demonstrate that the combined classifier achieves 95.9% accuracy, 96.1% sensitivity, and 95.0% specificity. An anonymized case study on empirical data ($N = 457$ actors) illustrates the protocol's application, without constituting a causal validation or generalization beyond the studied domain.

Contents

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1 Introduction

Low-dimensional visualization of semantic latent spaces—whether derived from embeddings, influence networks, or relational systems—is a critical tool for interpretation and decision-making. However, these visual maps typically lack formal geometric fidelity guarantees. Previous work by the author (Abella, 2026) has shown that embedding projections can collapse relevant interpretative distinctions; this work addresses the complementary problem of how to formally audit whether a 2D projection preserves the structure of the original latent space before interpreting it.

The credibility gap is direct: if 2D visualizations are not mathematically audited, they are reduced to subjective aesthetic representations. The protocol we present is designed to bridge this gap. *M15 does not certify the truth of the underlying model. It audits whether a 2D latent projection preserves sufficient metric, topological, and local semantic-structural integrity to support responsible interpretation.*

Existing literature on the quality evaluation of dimensional projections has traditionally focused on neighborhood preservation metrics such as *trustworthiness* and *continuity* (Venna & Kaski, 2001), stress measures (Kruskal, 1964; Shepard, 1962), and co-ranking analysis (Lee & Verleysen, 2007). These metrics evaluate the local preservation of neighborhoods but are insensitive to macroscopic deformations (collapses of topological bridges, cluster swaps) and local semantic-structural dissonances. Recently, persistent homology has been applied to evaluate dimensional reductions (Rieck & Leitte, 2015; Doraiswamy et al., 2021), and graph spectral signatures have been used to compare topologies (Tsitsulin et al., 2018; Sun et al., 2009). TopoMap and NetLSD represent relevant approaches to topological preservation and spectral graph comparison. However, M15 differs in that it does not propose a new layout algorithm or a global graph distance, but a post-hoc ordinal audit of an already generated 2D projection, combining metric fidelity, a PH0/MST profile, and interpretable local dissonance. In the domain of LLM judge evaluation, frameworks such as RULERS (Hong et al., 2026) address calibration and drift mitigation, while in graph anomaly detection, works like ComGA (Luo et al., 2022) and surveys by Ma et al. (2021) analyze structural anomaly identification. Additionally, CO-EVOLVE (Xing & Shahzad, 2026) addresses the semantic and structural co-evolution in heterophilous learning. However, no existing protocol simultaneously integrates global metric fidelity, hierarchical topological stability, and local semantic-structural dissonance detection into a unified audit framework.

It is worth noting the terminological origin of the protocol: the acronym *M15* directly refers to its engineering designation within the TRACE audit suite as *Motor 15* (Engine 15). In terms of practical development, M15 originally arose in the context of an exploratory case study of institutional influence, where the need to evaluate the consistency of a visual mapping and its local dissonances prompted the methodological coupling with a topological stability study. In this way, M15 is formulated as a post-hoc audit layer: it does not generate the projection or certify the truth of the underlying latent space, but rather evaluates whether the 2D representation preserves sufficient metric, topological, and local properties for responsible interpretation.

Although the exploratory case study focuses on an anonymized institutional ecosystem, M15 does not depend on the political domain. Its formal input is a latent matrix $X \in \mathbb{R}^{N\times D}$, a projection $Y \in \mathbb{R}^{N\times 2}$, and, optionally, a relational graph G. Therefore, the protocol can be applied to semantic embeddings, actor maps, influence networks, social graphs, or corporate spaces, as long as there is a 2D projection whose fidelity needs to be audited.

This paper proposes and evaluates this integrated protocol.

2 Methodological Framework

The protocol implements three complementary analysis layers to evaluate the projection at global, hierarchical, and local scales. The architecture of the audit flow is schematized in Figure 1.

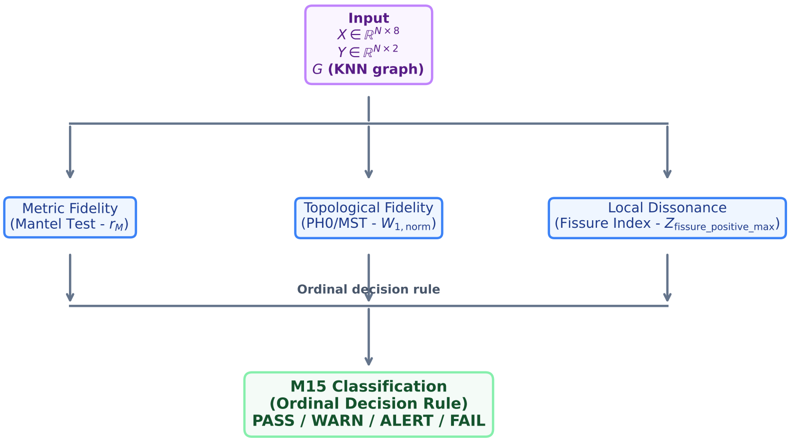

Figure 1: Flowchart of the M15 multiscale geometric audit protocol. Based on the original latent space, graph, and 2D projection, the global metric fidelity, hierarchical topology, and local dissonance are calculated in parallel, which are then aggregated to classify the state of the projection.

2.1 Mantel Test (Global Metric Fidelity)

Given two normalized distance matrices of size $N \times N$, D_X (from the original latent space) and D_Y (from the 2D projection), we calculate the Pearson correlation coefficient on the elements of the upper triangular part:

$$r_M = \frac{\sum_{i<j}(D_{X,ij} - \bar{D}_X)(D_{Y,ij} - \bar{D}_Y)}{\sqrt{\sum_{i<j}(D_{X,ij} - \bar{D}_X)^2 \sum_{i<j}(D_{Y,ij} - \bar{D}_Y)^2}}$$

To calculate statistical significance against the structural dependency of rows and columns, we perform a Monte Carlo shuffle, permuting the rows and columns of D_Y jointly 1,000 times to compute the empirical p-value of non-correlation (Mantel, 1967).

2.2 PH0 Profile over MST (Hierarchical Topological Stability)

0-dimensional persistent homology (PH_0) evaluates the hierarchical clustering of the representation. The edge weights of the Minimum Spanning Tree (MST) correspond to the death times of the connected components under Vietoris-Rips filtration (Cohen-Steiner et al., 2007; Edelsbrunner & Harer, 2010). The 2D metric compaction artificially alters the raw Wasserstein-1 distance, so we normalize the edges of both MSTs (E_X and E_Y) by their corresponding mean before calculating the normalized Wasserstein-1 distance:

$$W_{1,\text{norm}} = W_1\left(\frac{E_X}{\mu(E_X)}, \frac{E_Y}{\mu(E_Y)}\right)$$

The profile is complemented by the *MST total length ratio* ($R_{\text{MST}} = \sum E_Y / \sum E_X$), which measures overall metric compression, and the variance ratio ($V_{\text{ratio}} = |\log(\text{Var}(E_{Y,\text{norm}})/\text{Var}(E_{X,\text{norm}}))|$) to capture losses of topological heterogeneity. The formal connection between MST edge weights and 0D persistence is well established in the TDA literature (Rieck & Leitte, 2015; Doraiswamy et al., 2021).

2.3 Fissure Index (Local Semantic-Structural Dissonance)

To detect local misalignments between a node's position in the network (its structural community) and its semantic profile in the latent space, we calculate the Mahalanobis distance of each node with respect to its structural centroid and compare it with the distance to the closest alternative semantic centroid. The *Mahalanobis delta* ($\Delta_{\text{Mah}} = d_{\text{structural}} - d_{\text{alt}}$) quantifies the dissonance: high positive values indicate nodes whose semantic profile diverges from their structural community.

We generate contrasted Z-scores using a Monte Carlo null model with community-restricted permutation inspired by degree-corrected null models of the Karrer-Newman type: we permute the semantic vectors between nodes while keeping the community membership fixed ($N_{\text{perm}} = 1{,}000$). The Z-score of the Fissure Index depends on the permutation seed and the number of permutations; therefore, the values reported in Study 1 (Section 4) represent the mean over 50 independent runs accompanied by 95% bootstrap confidence intervals.

Unlike AGAD approaches based on neural reconstruction models or end-to-end learning, the Fissure Index is formulated as an interpretable post-hoc statistic: it evaluates structural-semantic dissonance using Mahalanobis distance and a community-restricted null model without training a deep detector.

Although the main diagnostic architecture of M15 is organized into three layers—Mantel, PH0/MST, and the Fissure Index—, the classifier's implementation retains the *Heat Trace Divergence* (D_{HT}) as an auxiliary spectral consistency check. Since D_{HT} did not provide primary discriminative power in the reported benchmark ($N = 100$), its results are documented in Appendix B and are not presented as a central contribution.

2.4 M15 Protocol Algorithm

To facilitate the reproducibility and implementation of the protocol, the general multiscale audit algorithm is described in Table 1.

2.5 Diagnostic Thresholds of the Ordinal Classifier

To ensure reproducibility and the practical implementation of M15, Table 2 defines the guideline thresholds used to determine the diagnostic category of the projection.

The rules are applied in descending order of severity: extreme `FAIL` conditions are evaluated first, followed by `ALERT`, then `WARN`, and finally `PASS`. When multiple conditions are triggered simultaneously, the most severe category prevails. The table summarizes the main diagnostic conditions; the implementation preserves D_{HT} as an auxiliary consistency check in the ordinal rule, documented in Appendix B. The thresholds are reported for reproducibility of the main benchmark. They should not be interpreted as universal scale invariants; Appendix C shows that some absolute thresholds require calibration by graph size or family.

3 Quantitative Evaluation via Synthetic Benchmark

3.1 Experimental Design

To validate the discriminative capability of the protocol against complex distortions, we designed an adversarial benchmark of 10 scenarios evaluated using $N_{\text{sim}} = 100$ Monte Carlo simulations. Each simulation generates an 8D synthetic latent space ($N = 100$ nodes) structured into two main communities separated in the dominant dimension, with a KNN graph ($k = 5$) built on the normalized Euclidean distances.

The 10 scenarios are:

Table 1: M15 Geometric Audit Protocol Algorithm

Algorithm 1: M15 Multiscale Geometric Audit Protocol

Input:

- Coordinate matrix of the original latent space: $X \in \mathbb{R}^{N \times D}$ (e.g. $D = 8$).
- Coordinate matrix of the 2D projection: $Y \in \mathbb{R}^{N \times 2}$ (e.g. FA2 or UMAP).
- Underlying relational or neighborhood graph: $G = (V, E)$.

Steps:

1. **Global Metric Layer (Mantel Test):**
   a. Compute normalized Euclidean distance matrices D_X (on X) and D_Y (on Y).
   b. Compute the Mantel correlation coefficient r_M between the upper triangular parts of D_X and D_Y.
   c. Perform $N_{\text{perm}} = 1{,}000$ joint permutations of rows and columns of D_Y to obtain the empirical p-value.
2. **Hierarchical Topological Layer (PH0 Profile/MST):**
   a. Compute Minimum Spanning Trees MST_X and MST_Y on D_X and D_Y.
   b. Normalize edge weights by their respective means: $E_{X,\text{norm}} = E_X/\mu(E_X)$, $E_{Y,\text{norm}} = E_Y/\mu(E_Y)$.
   c. Compute normalized Wasserstein-1 distance: $W_{1,\text{norm}} = W_1(E_{X,\text{norm}}, E_{Y,\text{norm}})$.
   d. Compute auxiliary topological descriptors: $R_{\text{MST}} = \sum E_Y / \sum E_X$ and V_{ratio}.
3. **Local Semantic-Structural Layer (Fissure Index):**
   a. For each node $i \in V$, compute the Mahalanobis dissonance: $\Delta_{\text{Mah}} = d_{\text{structural}} - d_{\text{alt}}$.
   b. Contrast Δ_{Mah} via community-restricted permutation inspired by degree-corrected Karrer-Newman null models to obtain the empirical Z-score of semantic dissonance.
   c. Estimate the 95% bootstrap confidence intervals of the Z-score using $n = 50$ independent replicates.
4. **Audit Classification (Ordinal Logic):**
   a. Classify the layout status: `PASS` (coherent), `WARN` (mild dissonance), `ALERT` (local anomalies/misalignment), or `FAIL` (severe distortion).

Output: Formal geometric audit report containing r_M, $W_{1,\text{norm}}$, $Z_{\text{fissure,positive,max}}$, and the classification.

Table 2: Guideline thresholds for the ordinal audit classification in M15.

Status	Guideline Condition	Interpretation
`PASS`	$r_M > 0.85$ and $W_{1,\text{norm}} < 0.55$ and $Z_{\text{fissure}} < 2.0$	Interpretable geometric projection
`WARN`	$0.70 < r_M \leq 0.85$ or $0.55 \leq W_{1,\text{norm}} < 0.70$ or $2.0 \leq Z_{\text{fissure}} < 3.0$	Moderate geometric distortion or mild dissonance
`ALERT`	$r_M \leq 0.70$ or $W_{1,\text{norm}} \geq 0.70$ or $Z_{\text{fissure}} \geq 3.0$	Relevant distortion or local dissonance
`FAIL`	$r_M < 0.30$ or $W_{1,\text{norm}} > 1.0$ or extreme combined condition	Unreliable projection for interpretation

1. **A_CLEAN:** Spring Layout projection (Fruchterman-Reingold force-directed algorithm in NetworkX) converged (180 iterations). Clean control with low distortion.
2. **B1_LOW_NOISE:** CLEAN projection + Gaussian noise $\sigma = 0.10$. Mild metric distortion.
3. **B2_MEDIUM_NOISE:** CLEAN projection + noise $\sigma = 0.35$. Moderate material distortion.
4. **B3_HIGH_NOISE:** CLEAN projection + noise $\sigma = 0.75$. Severe distortion.
5. **C_RANDOM:** Uniformly random 2D coordinates $U(-1, 1)$. Total structure destruction.
6. **D_BRIDGE_COLLAPSE:** Collapse of the inter-community bridge: one cluster overlaps the other in 2D, removing the visual separation. Designed to expose the blindness of *continuity* to macro collapses.
7. **E_FALSE_OUTLIER:** A central node is displaced to extreme coordinates $(5, 5)$ in 2D. Simulates layout artifacts. Designed to test the sensitivity of the PH0 Profile to the displacement of a single MST edge.
8. **F_REAL_FISSURE:** Semantic injection: a node in cluster 1 receives the 8D vector of a node in cluster 2, but retains its original 2D position. The only anomaly exclusively detectable by the Fissure Index.
9. **G_COMMUNITY_MISMATCH:** Exchange of 2D centroids between clusters, with 15% of nodes "stuck" in their original position (emulating layout convergence failure). Breaks metric isometry and generates local dissonance detectable by Fissure.
10. **H_SPECTRAL_REWIRE:** Rewiring of 15 inter-community edges in high dimension, followed by spectral embedding (Laplacian Eigenmaps 8D) of the rewired graph. Maintains the original 2D visualization. Destabilizes the global spectral topology.

Figure 2 shows a representative selection of adversarial scenarios, facilitating the understanding of the type of distortion evaluated.

3.2 Benchmark v2 Results (Means by Scenario)

Table 3 summarizes the means obtained for the main metrics of the protocol in each of the 10 evaluated scenarios.

Table 3: Results of the Benchmark v2 (Means ± Standard Deviation per Scenario over 100 simulations)

Scenario	Mantel r_M	$W_{1,\text{norm}}$	MST Ratio	$Z_{\text{fissure,positive,max}}$	Trust.	Cont.	Stress	Classif.
A_CLEAN	0.986 ± 0.001	0.481 ± 0.025	0.180 ± 0.008	0.00 ± 0.00	0.899 ± 0.010	0.953 ± 0.005	0.197 ± 0.008	PASS
B1_LOW_NOISE	0.973 ± 0.004	0.408 ± 0.040	0.203 ± 0.013	0.00 ± 0.00	0.825 ± 0.011	0.843 ± 0.013	0.231 ± 0.021	PASS
B2_MEDIUM_NOISE	0.819 ± 0.040	0.301 ± 0.044	0.315 ± 0.026	0.00 ± 0.00	0.780 ± 0.011	0.775 ± 0.014	0.419 ± 0.041	ALERT
B3_HIGH_NOISE	0.446 ± 0.083	0.286 ± 0.042	0.373 ± 0.029	0.00 ± 0.00	0.686 ± 0.030	0.662 ± 0.031	0.575 ± 0.039	FAIL
C_RANDOM	-0.001 ± 0.012	0.252 ± 0.033	0.413 ± 0.021	0.00 ± 0.00	0.508 ± 0.018	0.508 ± 0.018	0.632 ± 0.006	FAIL
D_BRIDGE_COLLAPSE	0.039 ± 0.008	0.217 ± 0.033	0.419 ± 0.029	0.00 ± 0.00	0.556 ± 0.023	0.813 ± 0.017	0.647 ± 0.013	FAIL
E_FALSE_OUTLIER	0.678 ± 0.038	1.232 ± 0.038	0.113 ± 0.009	0.00 ± 0.00	0.895 ± 0.011	0.940 ± 0.009	0.747 ± 0.024	FAIL
F_REAL_FISSURE	0.946 ± 0.003	0.482 ± 0.025	0.180 ± 0.008	3.57 ± 0.93	0.885 ± 0.010	0.939 ± 0.008	0.250 ± 0.006	ALERT
G_COMMUNITY_MISMATCH	0.479 ± 0.006	0.486 ± 0.029	0.177 ± 0.007	4.48 ± 0.68	0.729 ± 0.017	0.772 ± 0.018	0.591 ± 0.007	FAIL
H_SPECTRAL_REWIRE	0.324 ± 0.011	0.567 ± 0.023	0.154 ± 0.010	0.00 ± 0.00	0.871 ± 0.024	0.953 ± 0.008	0.636 ± 0.013	FAIL

The resulting classification distribution over the 100 Monte Carlo simulations for each scenario is represented in Figure 3.

Note that the D_{HT} (Heat Trace Divergence) column has been omitted from this table because it did not provide discrimination between scenarios (empirical range 0.011–0.015 in all 10 cases; see Appendix B).

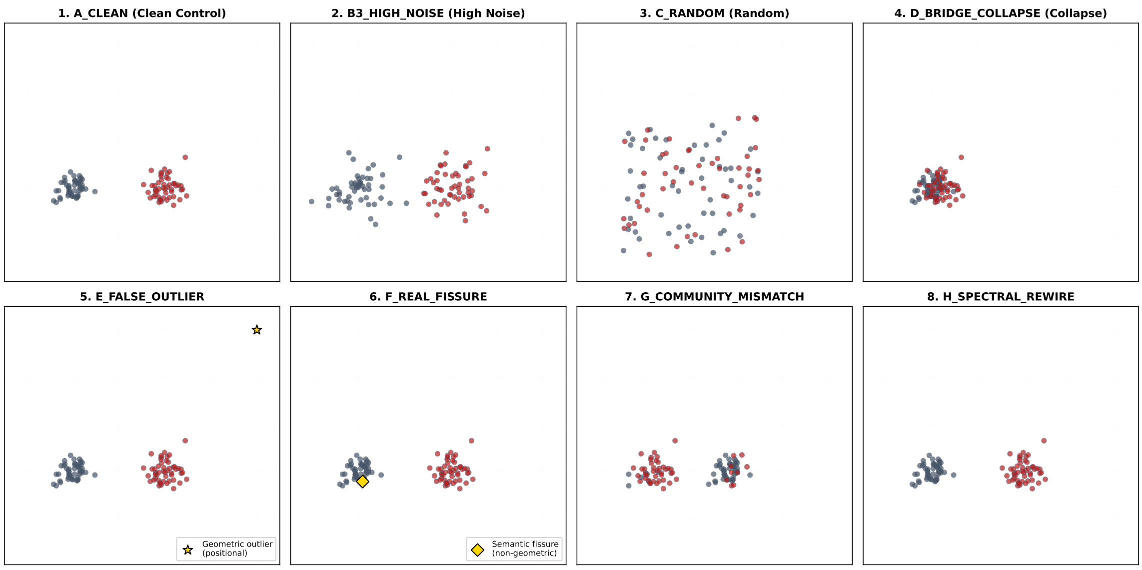

Figure 2: Representative scenarios of the adversarial benchmark ($N = 100$ nodes). Blue and red colors represent the two original high-dimensional structural communities. The metric, topological, and local dissonance anomalies introduced perturb the separability of the clusters or introduce anomalous nodes (gold stars/diamonds). Alteration H is not visually apparent in the 2D plane; it is introduced in the underlying connectivity/spectral structure.

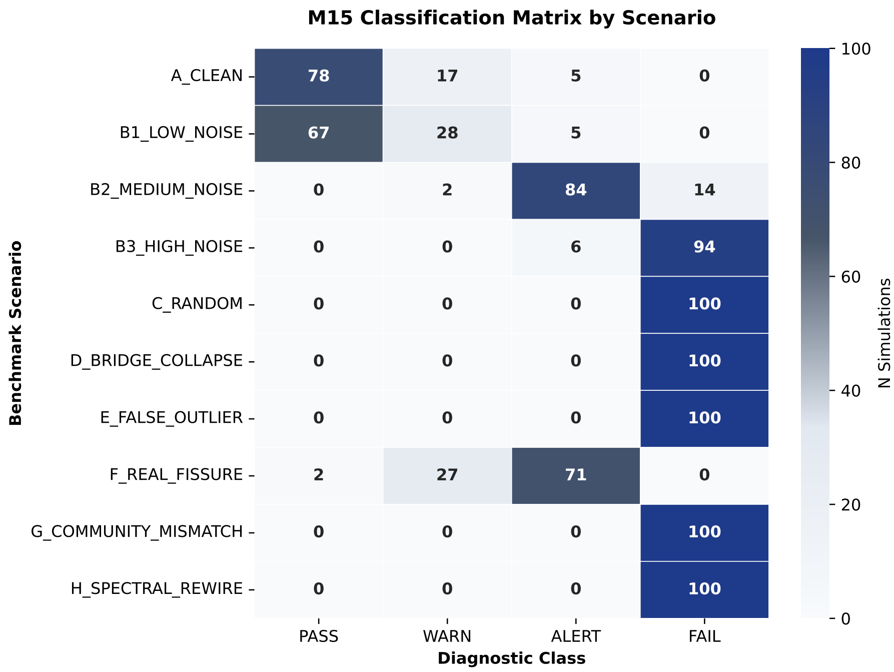

Figure 3: Classification heatmap of the M15 ordinal classifier over 100 Monte Carlo simulations per scenario. The figure shows the number of simulations classified in each diagnostic state over 100 runs per scenario (PASS, WARN, ALERT, FAIL), demonstrating the high sensitivity and specificity of the protocol.

The **D_BRIDGE_COLLAPSE** case is empirically central: *continuity* reaches 0.813 (an acceptable value according to literature criteria) in a scenario where the network is topologically broken. That is, a classical metric (Venna & Kaski, 2001) fails to detect the collapse of the bridge. The M15 protocol detects it via r_M (which drops to 0.039) and R_{MST} (which shoots up to 0.419).

On the other hand, Figure 5 illustrates the crucial diagnostic difference between an artificial positional displacement (Scenario E) and a local semantic fissure invisible at the macro level (Scenario F).

3.3 Ablation Study

To justify the necessity of the multiscale protocol, we compare the diagnostic accuracy of the complete model against its isolated components and against classical baselines from the literature. The binary evaluation classifies scenarios B2–H as "anomaly" (positive condition) and scenarios A_CLEAN and B1_LOW_NOISE as "OK" (negative condition). The results are detailed in Table 4.

Table 4: Ablation Study

Model	Accuracy	TPR	TNR	TP / TN / FP / FN
Trustworthiness-only	61.80%	64.62%	50.50%	517 / 101 / 99 / 283
Continuity-only	62.60%	62.12%	64.50%	497 / 129 / 71 / 303
Classical Comb. T+C (Venna & Kaski, 2001)	61.80%	64.62%	50.50%	517 / 101 / 99 / 283
Stress-only (Kruskal, 1964)	89.60%	87.00%	100.00%	696 / 200 / 0 / 104
Mantel-only	80.00%	75.00%	100.00%	600 / 200 / 0 / 200
PH0-only (MST TDA)	69.80%	62.25%	100.00%	498 / 200 / 0 / 302
Fissure-only	37.10%	21.38%	100.00%	171 / 200 / 0 / 629
Combined M15	**95.90%**	**96.12%**	**95.00%**	**769 / 190 / 10 / 31**

The visual comparison of the performance of each metric and the combined classifier is illustrated in Figure 4.

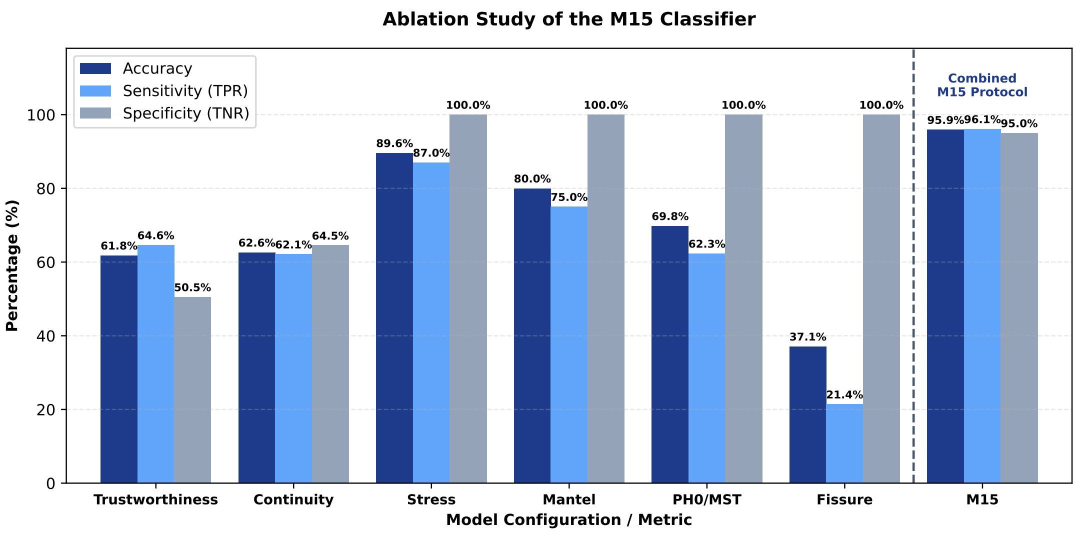

Figure 4: Comparison of diagnostic accuracy, sensitivity (TPR), and specificity (TNR) in the ablation study. The combined M15 multiscale classifier consistently outperforms all individual baselines and combinations from the literature. Stress achieves high specificity but lower balanced detection than M15.

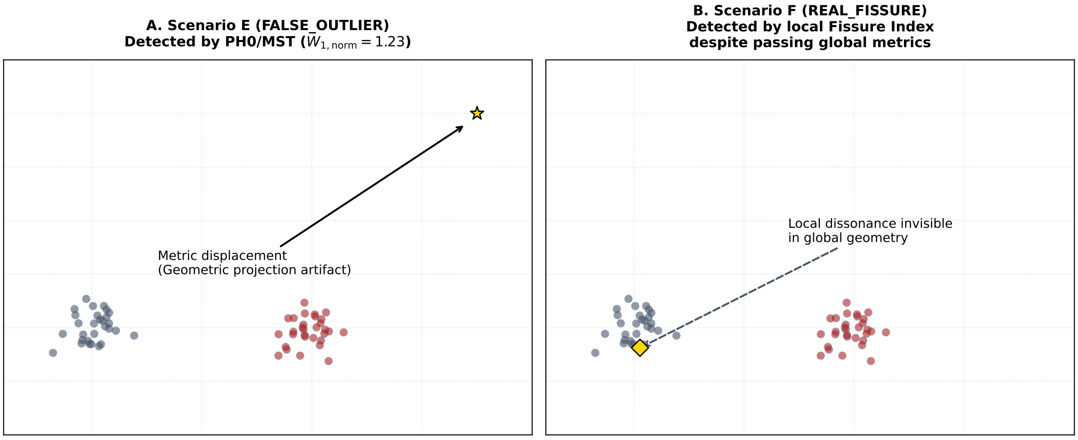

Figure 5: Comparison between geometric displacement (A) and local semantic fissure (B). In E, the metric projection is deformed by moving a node away, detected by the length of the MST ($W_{1,\mathrm{norm}}$). In F, the positional layout in 2D is apparently clean, but the node is semantically dissonant with its relational community, being detected solely by the Fissure Index.

The empirical synergy is clear: Mantel alone achieves 80% because it is blind to local distortions (E, F). PH0 alone achieves 69.8% because it does not capture pure semantic dissonance. Fissure alone reaches only 37.1% because it is completely blind to global disorder. The combination reaches 95.9%, with the strongest gain coming from the interaction of the PH0/MST + Fissure Index.

It is worth noting that while the model based solely on Stress (*Stress-only*) shows robust performance with 100.0% specificity and 89.60% accuracy, its sensitivity (TPR = 87.00%) is noticeably lower than that of the combined M15 protocol (96.12%). This shows that analyzing only global metric stress tends to underestimate or omit critical local distortions (such as local semantic fissures in Scenario F), fully justifying the need to integrate multiple layers in the proposed approach.

The calibration of the ordinal classifier prioritizes specificity over maximum local sensitivity: under the current strict threshold, the local fissure condition is detected in 71% of the simulations of scenario F, without catastrophic false positives in clean projections (0 cases of A_CLEAN classified as FAIL).

4 Exploratory Case Study on Anonymized Data ($N = 457$)

To illustrate the practical application of the protocol, we perform an audit on an anonymized empirical dataset of $N = 457$ actors from an institutional ecosystem vectorized in an 8-dimensional latent space (CRED, CAP, LEAD, PROX, INTG, POL, COH, SAL).

The empirical case study is included as an anonymized illustrative study. The original unmasked data are protected by a non-disclosure agreement (NDA) and cannot be shared publicly. The case does not constitute the primary validation of M15, and no causal claim is made about identifiable actors, organizations, or infrastructures. Organization labels, actor identities, and OSINT operational traces are masked to prevent re-identification. The primary validation of the protocol rests on the synthetic adversarial benchmark and the ablation study.

The 8D latent space was projected to 2D using the ForceAtlas2 continuous force-directed layout algorithm (implemented in Gephi, 100 iterations) and UMAP non-linear dimensionality reduction ($k = 15$, $d_{\min} = 0.1$).

4.1 Metric Fidelity and Topological Stability Results

Table 5 shows the results of the global metrics applied to the empirical case.

Table 5: Results of Metric Fidelity and Topological Stability in the Empirical Case

Metric	ForceAtlas2 Layout	UMAP Layout
Mantel r_M	0.7456 ($p < 0.0001$)	0.3137 ($p < 0.0001$)
Wasserstein-1 (W_1, 0D)	0.0294	0.0368

Both projections show a statistically significant metric association, although with clearly different magnitudes. The ForceAtlas2 layout preserves the global distance relationship better than UMAP ($r_M = 0.75$ vs. 0.31), so it is used solely as an exploratory reference visualization for macroscopic blocks. Wasserstein-1 distances for 0D persistence are low in both cases, suggesting the absence of severe topological fragmentation under this metric. These results do not imply causal validation or substantive authorization of the empirical map. Figure :layout$_c omparisonvisuallyillustratesthearrangementofactorsinbothlayouts, coloredaccordingtotheanony$

4.2 Fissure Index Results (Bootstrap, $n = 50$ seeds)

The Z-scores of the Fissure Index are sensitive to the seed of the permutation null model. To guarantee reproducibility, we ran the full calculation 50 times with independent seeds (0–49), each with $N_{\text{perm}} = 1{,}000$ community-restricted permutations inspired by degree-corrected Karrer-Newman null models. The reported values are **mean ± standard deviation**, accompanied by **95% bootstrap confidence intervals** (2.5th and 97.5th percentiles). The observed inter-seed variance is low ($\sigma < 0.12$ for all focal actors), confirming the stability of the estimator under the null model used.

Table 6 presents the focal actors with the highest positive dissonance (LOCAL ALERT) and the main node of high negative fidelity, using anonymous identifiers in accordance with the confidentiality scheme (where focal actors are denoted as Actor F1–F5 and H1):

Table 6: Fissure Index Results (Bootstrap, $n = 50$ seeds)

Actor	$\bar{Z}$	σ_Z	95% CI	Stable ($Z > 3.0$)?
Actor F1	+4.83	0.12	[+4.59, +5.02]	**Yes**
Actor F2	+4.23	0.12	[+4.00, +4.46]	**Yes**
Actor F3	+3.24	0.12	[+3.05, +3.43]	**Yes**
Actor F4	+2.97	0.06	[+2.85, +3.08]	Borderline (crosses 3.0)
Actor F5	+2.64	0.09	[+2.50, +2.77]	No (CI < 3.0)
Actor H1	−2.46	0.08	[−2.60, −2.33]	**Stable negative**

Interpretation:

- **Three actors in a stable LOCAL ALERT state ($Z > 3.0$):** Actor F1 ($\bar{Z} = +4.83$), Actor F2 ($\bar{Z} = +4.23$), and Actor F3 ($\bar{Z} = +3.24$) exhibit statistically significant and stable semantic-structural dissonance under bootstrap. Their semantic profile diverges from their assigned structural community, which is compatible with a local dissonance between their semantic profile and community embedding, whose causal interpretation requires external domain analysis.

- **Two actors with suggestive but unconfirmed signal:** Actor F4 ($\bar{Z} = +2.97$, CI crosses $Z = 3.0$) and Actor F5 ($\bar{Z} = +2.64$, CI completely below 3.0). They are reported as moderate anomalies whose significance does not reach the strict threshold under bootstrap.
- **Actor H1 — Low dissonance hub:** Actor H1 presents a stable negative Z-score ($\bar{Z} = -2.46$, CI $[-2.60, -2.33]$), compatible with low local dissonance between its semantic profile and its community embedding under the null model used. The value previously reported in the preliminary report ($Z = -3.83$) falls outside the bootstrap CI and is attributed to seed variance in a single run. *(Note: the dataset contains a duplicate record 'Actor H1' / 'Actor H1-alt' in different clusters; the reported primary value corresponds to the cluster 0 record, $\bar{Z} = -2.46$.)*
- **Global bootstrap statistics:** Out of the 457 actors, 399 (87.3%) present negatively shifted Z-scores under the null model used, suggesting low local dissonance according to this metric. This reading should not be interpreted as a substantive validation of the actors or their real relationships. Only 3 actors (0.7%) exceed the $Z > 3.0$ threshold in a stable manner.

Figure 6 presents the forest plot of the Fissure Index Z-scores for the six focal actors analyzed, showing the robustness and stability of the Z-scores.

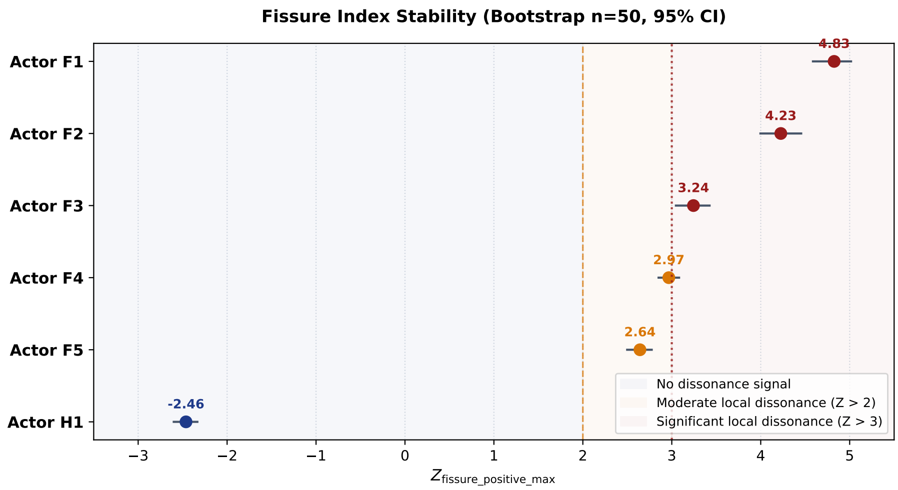

Figure 6: Forest plot of the Fissure Index Z-scores for the six focal actors in the anonymized case study. The horizontal error bars indicate the 95% confidence intervals calculated using $n = 50$ independent bootstrap replicates. The red dashed vertical line indicates the strict significance threshold $Z = 3.0$ for significant semantic-structural dissonance (LOCAL ALERT).

5 Discussion

5.1 Contribution and Scientific Claim

The synthetic benchmark v2 provides evidence that the M15 protocol improves projection validation compared to individual metrics in the literature by combining global metric fidelity (Mantel Test), hierarchical topological stability (PH0 Profile/MST), and local semantic-structural dissonance detection (Fissure Index). The diagnostic accuracy of the combined model (95.9%) exceeds the classical neighborhood baseline T+C (61.8%) and each isolated component in the benchmark.

The **D_BRIDGE_COLLAPSE** case is particularly relevant: it demonstrates empirically that a widely used metric in the literature (*continuity* $= 0.813$) overlooks a macroscopic topological

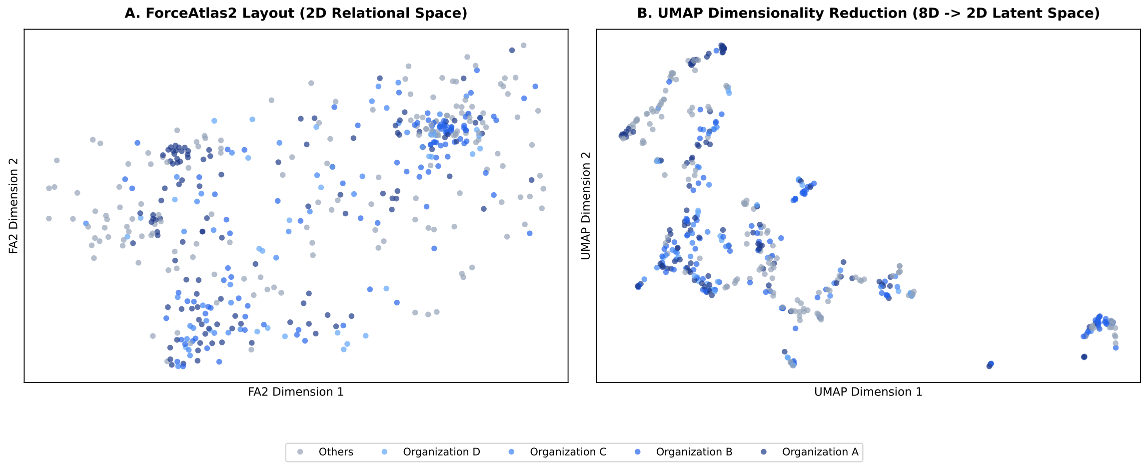

Figure 7: Comparison of 2D projections for the anonymized case study ($N = 457$ actors). (A) Relational layout obtained via the ForceAtlas2 continuous force-directed algorithm. (B) Layout obtained through UMAP non-linear dimensionality reduction applied to the 8-dimensional latent space. Nodes are colored according to their anonymously assigned organization. The empirical design is shown solely as an anonymized exploratory case study; no causal interpretation is inferred.

collapse that the M15 protocol detects unambiguously. This single scenario alone justifies the need for a framework that transcends local neighborhood metrics.

The protocol is applicable to auditing dimensional representations in contexts where the geometric fidelity of the 2D visualization is relevant for subsequent interpretation: forensic analysis, technical assessment, institutional compliance evaluation, or any domain where the decision depends on the visual mapping of a latent space. Although the illustrative case focuses on an anonymized institutional ecosystem, M15 does not depend on the political domain. Its formal input is a latent matrix $X \in \mathbb{R}^{N \times D}$, a projection $Y \in \mathbb{R}^{N \times 2}$, and, optionally, a relational graph G. Therefore, the protocol can be applied to semantic embeddings, actor maps, influence networks, social graphs, or corporate spaces, as long as there is a 2D projection whose fidelity needs to be audited.

Likewise, the scale dependency identified in absolute topological metrics does not invalidate the methodological proposal, but rather establishes limits on the direct transfer of fixed thresholds. This limitation is identified, quantified, and a preliminary operational calibration pathway exists through scale references and the complementary use of the most robust signals (r_M and the Fissure Index).

5.2 Limitations

1. **Scale dependency of topological metrics.** Scale dependency is not interpreted as a refutation of the protocol, but as an operational constraint on absolute thresholds. Supplementary results show that topological metrics such as $W_{1,\mathrm{norm}}$ and R_{MST} vary systematically with graph size even under clean conditions. Therefore, these values should not be used as universal thresholds outside the calibrated range. This limitation is identified, quantified, and a preliminary operational calibration pathway by scale reference is detailed in Appendix C. In the absence of specific calibration, the conservative use of the most robust signals (r_M and Fissure Index) is proposed.

2. **Risk of methodological overfitting.** The 10 adversarial scenarios were designed by the same team that developed the protocol. An adversarial benchmark designed by independent researchers would provide a more robust test.

3. **Heat Trace Divergence.** As documented in Appendix B, D_{HT} did not reach primary discriminative power in the benchmark range ($N = 100$). Its potential utility under larger graphs or more extreme spectral scenarios remains as future research.

4. **Fissure Index stability.** The Fissure Index Z-score depends on the seed and the number of permutations of the null model. Bootstrap confidence intervals are reported, but evaluation under alternative null models (DCSBM with overlap, configuration model) remains to be performed.

5. **External validation with drift ground truth.** Application to public datasets (Iris, Digits) demonstrates consistency with expected diagnoses from the literature on PCA/t-SNE/UMAP, but these datasets do not contain semantic-structural drift ground truth. An external validation on a dataset with known and verifiable semantic fissures would materially strengthen the protocol's empirical claim.

6. **Study 1 sample.** The anonymized empirical case study ($N = 457$) provides an applied illustration, not an independent statistical test of the protocol. Generalization to other domains requires additional replication.

7. **Verifiability of the empirical case.** The anonymized empirical case study is not externally verifiable by readers without access under NDA; therefore, it is reported as an applied illustration and not as the primary validation. Consequently, the empirical case study is not used to estimate the diagnostic performance of the protocol.

5.3 Computational Complexity and Performance

To evaluate the viability of the protocol in practical audit workflows, the computational cost of the M15 analysis pipeline was estimated. The dominant algorithmic complexity is determined by the Monte Carlo simulation of the Mantel Test ($O(N^2 \cdot N_{\text{perm}})$) and the restricted permutations of the Fissure Index ($O(N \cdot N_{\text{perm}} \cdot D)$). On a standard single-core consumer CPU (2.5 GHz), auditing a graph with $N = 100$ nodes takes less than 0.2 seconds of execution time. The complete execution of the $1,000$ simulations of the synthetic benchmark (10 scenarios, 100 Monte Carlo simulations per scenario with $N_{\text{perm}} = 1,000$) required a total compute time of approximately 3.2 minutes. The results suggest computational feasibility for moderate-scale audits. Performance on larger graphs will depend on the number of permutations, the distance calculation implementation, and the hardware architecture used.

The empirical case study is reported solely as an anonymized applied illustration. No substantive comparisons between organizations or causal or political inferences are included, in order to reduce the risk of contextual re-identification.

6 Conclusions

We present a multiscale post-hoc geometric audit protocol for latent space projections that integrates three complementary layers: global metric fidelity (Mantel Test), hierarchical topological stability (PH0 Profile/MST), and local semantic-structural dissonance detection (Fissure Index). Evaluation on an adversarial synthetic benchmark of 10 scenarios (1,000 Monte Carlo simulations) demonstrates that the combination of the three layers achieves a diagnostic accuracy of 95.9%, consistently outperforming each isolated component and classical neighborhood metrics in the literature. An anonymized case study ($N = 457$ actors) illustrates the practical application of the protocol, without constituting a causal validation or generalization beyond the studied domain.

The code and benchmark data are available for reproduction.

Author Contribution Statement (CRediT)

Rubén Abella: Conceptualization (lead), Methodology (lead), Formal analysis (lead), Investigation (lead), Resources (co-lead), Data curation (co-lead), Writing - original draft (lead), Writing - review & editing (lead), Visualization (lead), Project administration (lead), Funding acquisition (lead).

José Picón: Resources (co-lead), Data curation (co-lead), Investigation (support), Validation (support), Writing - review & editing (support).

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A Study of Empirical Consistency in Public Datasets

As a consistency exercise (not a validation), the M15 protocol was applied to the public datasets **Iris** ($N = 150$, $D = 4$, 3 classes) and **Digits MNIST** ($N = 300$ downsampled, $D = 64$, 10 classes), auditing the projections generated by PCA, t-SNE (van der Maaten & Hinton, 2008), and UMAP (McInnes et al., 2018). The numerical results are detailed in Table 7.

Table 7: Empirical results of the M15 audit on public datasets under PCA, t-SNE, and UMAP projections. Local alerts ($Z > 2.0$ and $Z > 3.0$) were calculated via K-Means ($K = 3$ for Iris, $K = 10$ for Digits).

Dataset	Projection	Mantel r_M	MST Death (W_1)	Alerts ($Z > 2.0$)	Alerts ($Z > 3.0$)
Iris	PCA	0.9984	0.0190	5 (3.3%)	1 (0.7%)
Iris	t-SNE	0.9261	0.0315	9 (6.0%)	2 (1.3%)
Iris	UMAP	0.9333	0.0356	5 (3.3%)	0 (0.0%)
Digits (MNIST)	PCA	0.6238	0.2680	6 (2.0%)	1 (0.3%)
Digits (MNIST)	t-SNE	0.6096	0.2746	1 (0.3%)	0 (0.0%)
Digits (MNIST)	UMAP	0.3207	0.2895	0 (0.0%)	0 (0.0%)

The M15 diagnostics are consistent with the literature: PCA retains global metric fidelity (high Mantel) but generates local dissonance alerts due to neighborhood overlap; t-SNE isolates clusters but collapses global Mantel; UMAP offers an intermediate balance. These results demonstrate that M15 metrics produce diagnostics consistent with the known behaviors of these algorithms, but **do not constitute an external validation** in the strict sense, since neither dataset contains verifiable semantic-structural drift ground truth. Fissure Index alerts generated on K-Means clusters are not confirmable true positives.

B Heat Trace Divergence: Auxiliary Result

The protocol includes the *Heat Trace Divergence* (D_{HT}) as a complementary metric, based on the comparison of heat traces of normalized Laplacians in logarithmic diffusion scales (Sun et al., 2009; Tsitsulin et al., 2018):

$$D_{HT} = \sqrt{\frac{1}{|T|}\sum_{t \in T}\left(\frac{1}{N}\mathrm{Tr}(e^{-tL_X}) - \frac{1}{N}\mathrm{Tr}(e^{-tL_Y})\right)^2}$$

In the synthetic benchmark ($N = 100$, $N_{\mathrm{sim}} = 100$), D_{HT} did not achieve significant discriminative power between scenarios (empirical range 0.009–0.021, total overlap of confidence intervals between all scenarios, including CLEAN and RANDOM). Its operational role in the M15 classifier is that of a secondary spectral consistency check. The main empirical gains come from the PH0/MST profile and the local Fissure Index.

The implementation of the ordinal classifier preserves D_{HT} as an auxiliary consistency check, contributing to the thresholds of the WARN and ALERT categories. However, its marginal contribution to diagnostic performance is minimal in the parameter range explored. Future research should evaluate whether D_{HT} acquires discriminative power in larger graphs ($N > 500$) or under more extreme spectral scenarios.

C Scale Stress Test and Ordinal Robustness

The main results of the paper are based on the standard adversarial benchmark with $N = 100$, 10 scenarios, and 1,000 Monte Carlo simulations. That benchmark is not modified. However, a natural limitation of any ordinal classifier based on absolute thresholds is its possible sensitivity to graph size.

To explore this question, we perform a multi-scale stress test with $N \in \{50, 100, 200, 300, 500\}$ over the 10 adversarial scenarios of the main benchmark (5 simulations per combination, with 250 total evaluations). The objective is not to replace the main benchmark, but to evaluate whether the ordinal decisions PASS/WARN/ALERT/FAIL remain stable under scale changes.

The results show that absolute topological metrics, in particular $W_{1,\mathrm{norm}}$ and R_{MST}, are scale-dependent. Their expected values under the clean scenario change systematically with N, which degrades the accuracy of fixed thresholds when multiple scales are aggregated. This scale dependency is not interpreted as a refutation of the protocol, but as an operational constraint on absolute thresholds. Therefore, these values should not be used as universal thresholds outside the calibrated range.

To apply M15 to graphs of a different size—for example, $N = 500$, $N = 2,000$, or larger—a scale reference calibration is proposed. This consists of estimating, for the corresponding graph size or family, a clean reference distribution of the topological metrics and normalizing the observed values with respect to this baseline:

$$z_W = \frac{W_{1,\mathrm{norm}}^{\mathrm{obs}} - \mu_W(N)}{\sigma_W(N)} \tag{1}$$

$$z_R = \frac{R_{\mathrm{MST}}^{\mathrm{obs}} - \mu_R(N)}{\sigma_R(N)} \tag{2}$$

The ordinal decision PASS/WARN/ALERT/FAIL must then be formulated on metrics calibrated by scale and structural usability criteria, not on absolute values directly transferred from $N = 100$. To facilitate the application of the method, Table 8 provides the empirical calibration values calculated from the simulations of the clean scenario A_{CLEAN} for various network scales, serving as initial discrete calibration curves. These values should be interpreted as an initial reference, not as definitive calibration curves.

Table 8: Empirical reference values (μ and σ) for scale calibration under the clean control scenario (A_{CLEAN}) as a function of the number of nodes N (5 seeds per scale).

N	$\mu_W(N)$	$\sigma_W(N)$	$\mu_R(N)$	$\sigma_R(N)$
50	0.668	0.024	0.235	0.009
100	0.485	0.027	0.178	0.008
200	0.260	0.018	0.153	0.006
300	0.196	0.006	0.146	0.008
500	0.110	0.013	0.165	0.004

We also evaluated an adaptive normalization based on z-scores with respect to a clean baseline estimated for each scale. Although this approach partially corrects the scale dependency, it introduces a conceptual problem: it measures deviation with respect to a clean projection of low variance, not necessarily loss of structural usability. Consequently, low or medium noise scenarios can receive excessive severities simply because the variance of the clean baseline is very narrow. A relevant result of this supplementary analysis is that the operational question is not whether a projection differs from an ideal clean baseline, but whether it preserves sufficient structural fidelity to be interpretable. Normalization by z-score penalizes minor deviations from low-variance clean scenarios, whereas the combination of global metric fidelity and local Fissure Index aligns better with the ordinal notion of diagnostic usability. The scale limitation, therefore, is identified, quantified, and a preliminary operational calibration pathway exists.

In the absence of a specific calibration for the target size, M15 can be applied in a conservative mode using the most robust signals observed in the multi-scale stress test: global metric fidelity (r_M) and the local Fissure Index. This variant does not replace the complete three-layer architecture, but it avoids improperly extrapolating absolute topological thresholds.

The most robust result was obtained with a simplified rule based on global metric fidelity (r_M) and local dissonance via the Fissure Index. This variant achieved an exact accuracy of 84.0% and an adjacent ordinal accuracy of 99.6% in the 250 multi-scale evaluations. That is, almost all errors were limited to contiguous categories within the PASS/WARN/ALERT/FAIL scale, failing in a non-adjacent manner in only 1 of the 250 evaluations.

Table 9: Comparison of multi-scale ordinal robustness (N = 50–500, 10 scenarios, 5 seeds). Adjacent accuracy considers a prediction correct if it is at most one category away on the ordinal PASS/WARN/ALERT/FAIL scale.

Classifier	Exact accuracy	Adjacent accuracy (±1)
Fixed thresholds	64.0%	91.6%
Adaptive z-score	64.0%	90.0%
r_M + Fissure Index	84.0%	99.6%

These results suggest that the cross-scale robustness of M15 does not depend on assuming universal topological thresholds. In the multi-scale stress test, absolute topological metrics show scale dependency, while the combination of global metric fidelity (r_M) and local dissonance (Fissure Index) retains high ordinal robustness in this preliminary test. This does not eliminate the diagnostic value of PH_0/MST in the main benchmark, but it suggests that absolute topological thresholds require recalibration for cross-scale operational use.

This analysis should be interpreted as a complementary stress test, not as a replacement for the main benchmark. The M15 architecture retains its multiscale formulation in the body of the article; this appendix identifies an operational pathway for future scale-aware variants of the protocol.

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